

CPE301D – SPRING 2023

Design Assignment - KICAD

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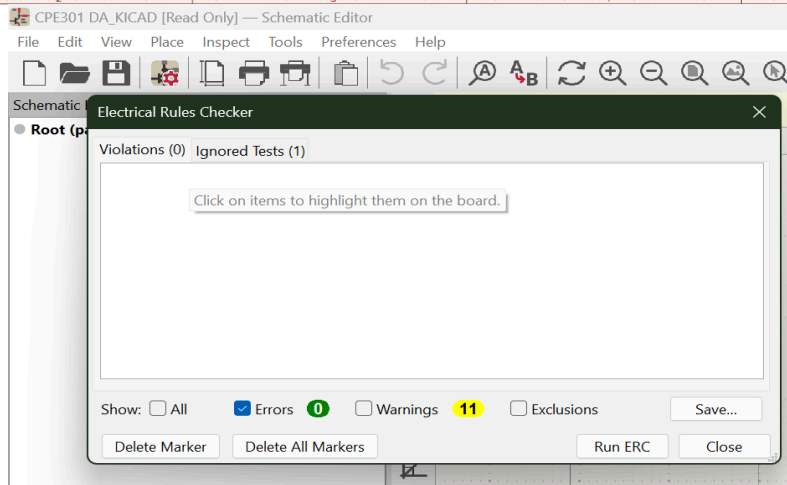
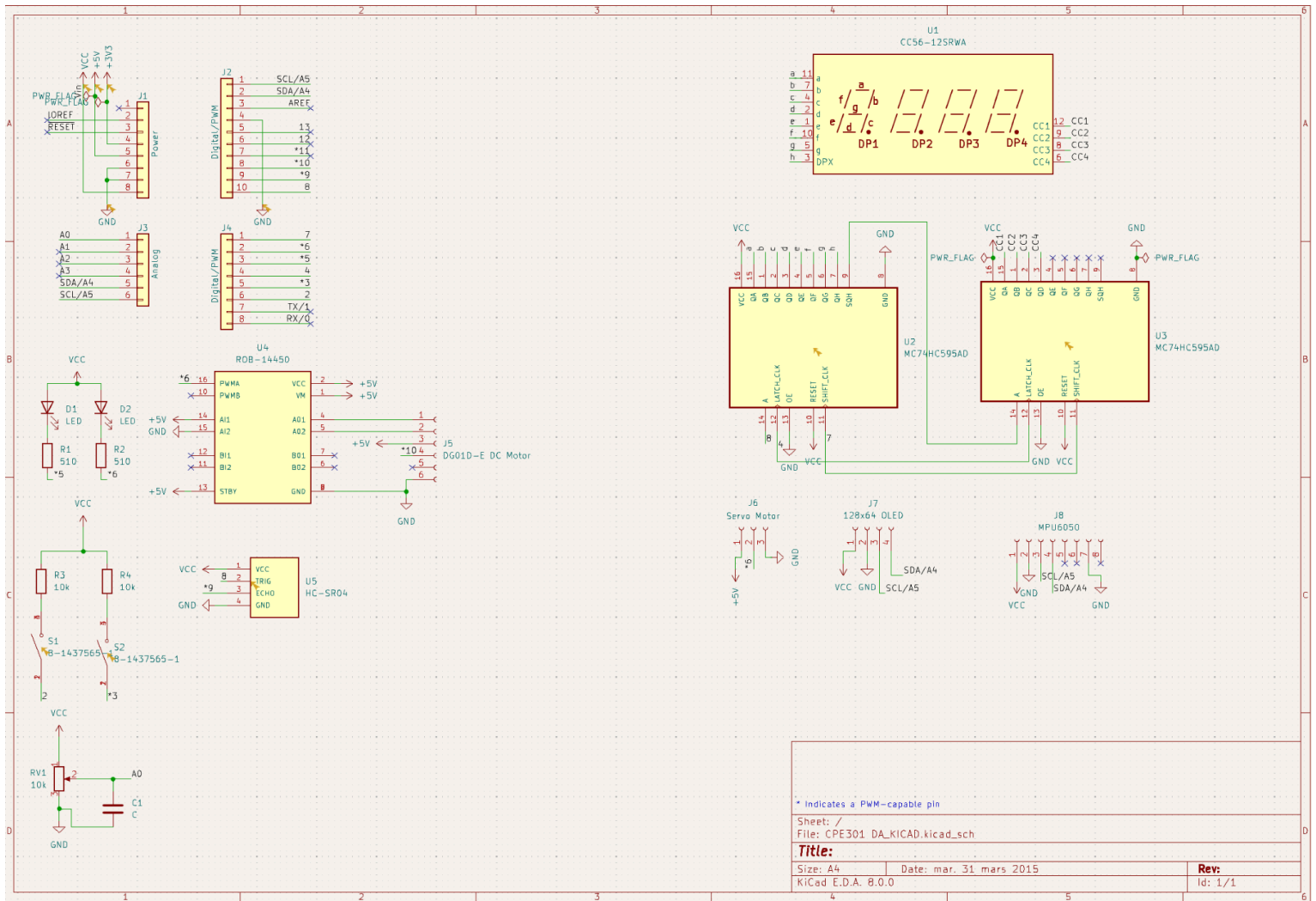
Directory: https://github.com/JoshuaMa2003/submission_da

1. COMPONENTS LIST

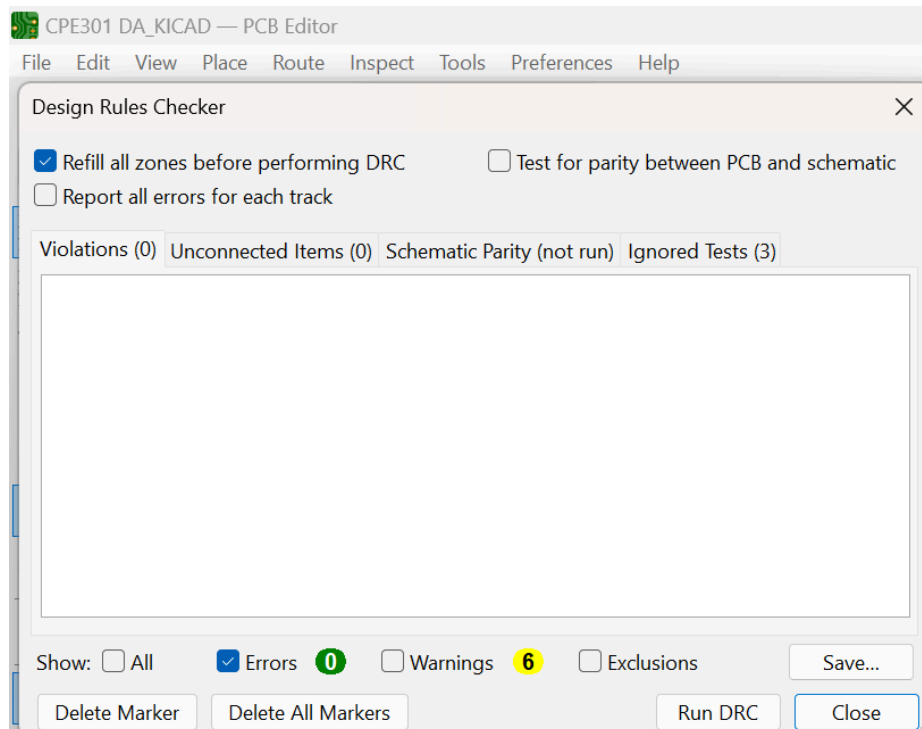
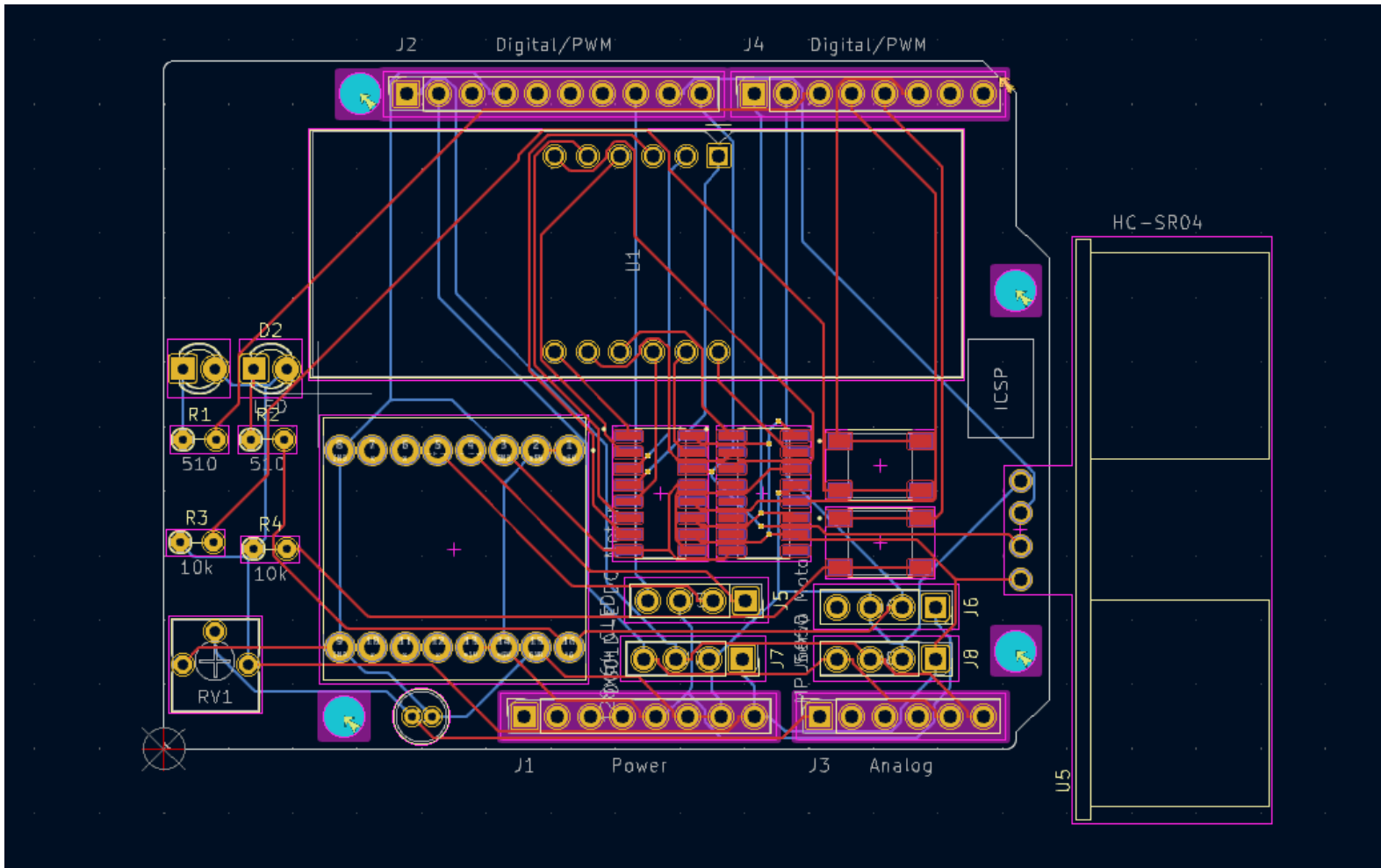
List of Components used

- **1 x Arduino Shield**
- **2 x LED**
- **2 x 510 Ω Resistors (for LEDs)**
- **2 x 8-1437565 Push Buttons**
- **2 x 10k Ω Resistors (for Push Buttons)**
- **1 x 10k Potentiometer**
- **1 x Capacitor (for Potentiometer)**
- **1 x ROB-14450 Motor Driver**
- **1 x HC-SR04 Ultrasonic Sensor**
- **1 x CC56-12SRWA Seven-Segment Display**
- **2 x MC74HC595AD Shift Registers**
- **1 x 1x6 Socket (DG01D-E Motor)**
- **1 x 1x3 Socket (Servo Motor)**
- **1 x 1x4 Socket (128x64 OLED)**
- **1 x 1x8 Socket (MPU6050 Sensor)**

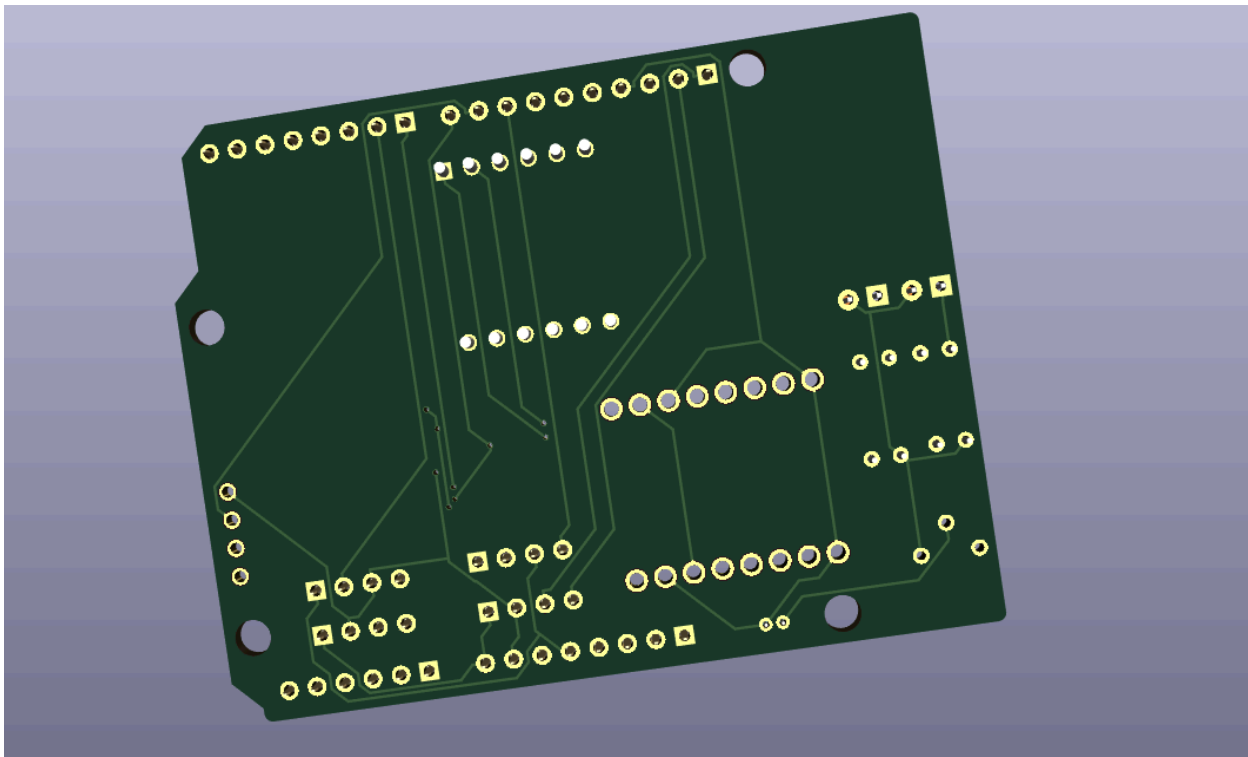
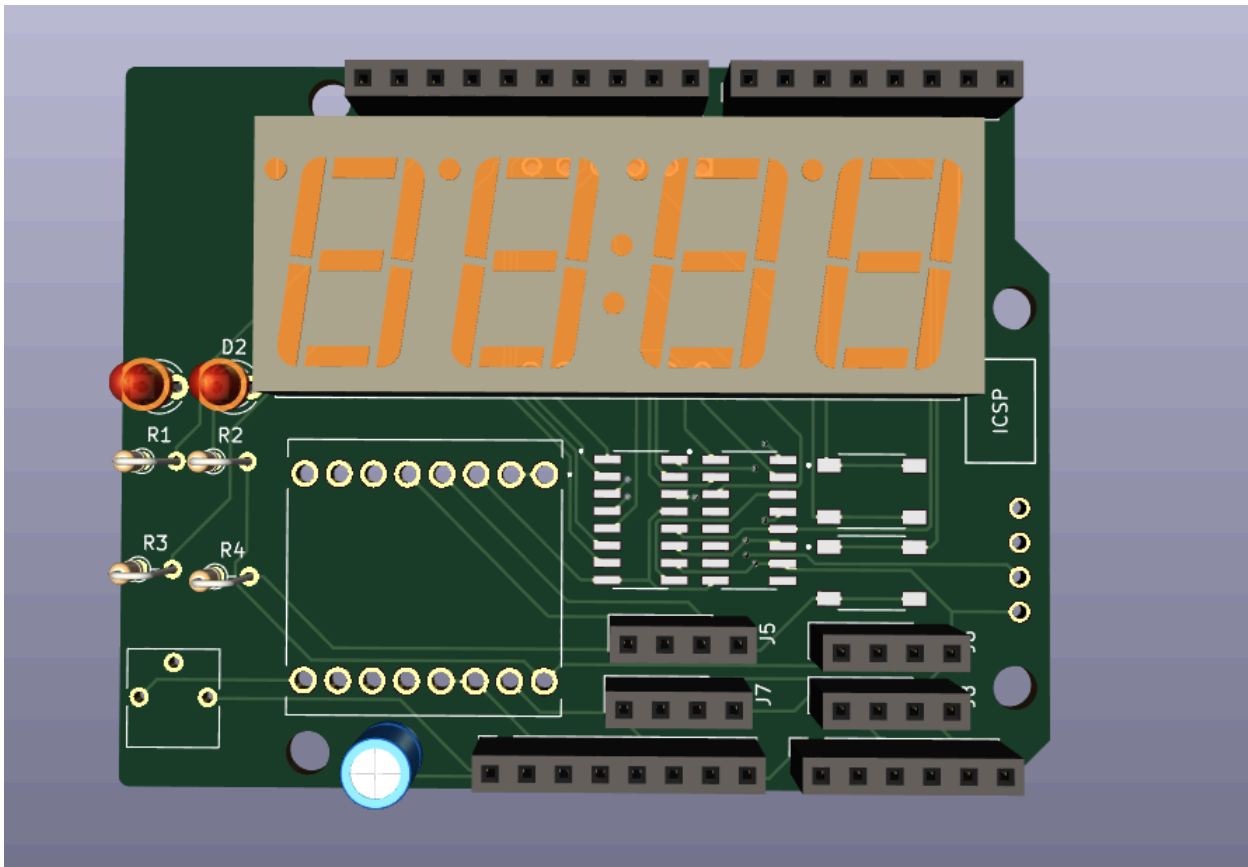
2. SCHEMATICS



3. LAYOUT



4. 3D VIEW (Front & Back)



5. BILL OF MATERIALS

Preview:

Reference	Value	Datasheet	Footprint	Qty	DNP
C1	C	~	Capacitor_THT:C_Radial_D4.0mm_H7.0mm_P1.50mm	1	
D1,D2	LED	~	LED_THT:LED_D3.0mm	2	
J1	Power	~	Connector_PinSocket_2.54mm:PinSocket_1x08_P2.54mm_Vertical	1	
J2,J4	Digital/FWM	-- mixed values --		2	
J3	Analog	~	Connector_PinSocket_2.54mm:PinSocket_1x06_P2.54mm_Vertical	1	
J5	DG01D-E DC Motor	~	Connector_PinSocket_2.54mm:PinSocket_1x04_P2.54mm_Vertical	1	
J6	Servo Motor	~	Connector_PinSocket_2.54mm:PinSocket_1x04_P2.54mm_Vertical	1	
J7	128x64 OLED	~	Connector_PinSocket_2.54mm:PinSocket_1x04_P2.54mm_Vertical	1	
J8	MPU6050	~	Connector_PinSocket_2.54mm:PinSocket_1x04_P2.54mm_Vertical	1	
R1,R2	510	~	Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm_P2.54mm_Vertical	2	
R3,R4,RV1	10k	~	-- mixed values --	3	
S1,S2	8-1437565-1	~	8-1437565-1:SW_8-1437565-1	2	
U1	CC56-12SRWA	http://www.kingbrightusa.com/images/catalog/SPEC/CC56-12SRWA.pdf	Display_7Segment:CA56-12SRWA	1	
U2,U3	MC74HC595AD	MC74HC595AD:SOIC127P600X175-16N	2		
U4	ROB-14450	ROB-14450:MODULE_ROB-14450	1		
U5	HC-SR04	HC-SR04:XCVR_HC-SR04	1		

Student Academic Misconduct Policy

<http://studentconduct.unlv.edu/misconduct/policy.html>

"This assignment submission is my own, original work".

JOSHUA MARTINEZ